RELATIONSHIP WORKSHOP

DSA0602 DATA HANDLING AND VISUALIZATION | Class Activity 2: Relationship Workshop in R (35 Marks) | Slot D

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Scatterplots | Correlograms | PCA | Networks | Paired data | Bubble charts | Hexbin plots | 7 tasks x 5 marks | Tool: R (ggplot2 and the packages named in each question)

Instructions: Tool: R (ggplot2 and the packages named). Time: 3 hours. Total: 35 marks (7 tasks x 5). Every student does the same seven tasks; the data in the R code printed with each task are your own, so your numbers will differ from your neighbour's. Run each code block unchanged (do not edit the seed). Draw the chart asked for, then answer with specific values from YOUR output; generic statements earn no credit. Each chart needs a title, labelled axes with units and colour-blind-safe colours. Submit one R script, your charts and short answers as one file named RollNo_Name.

Q1. Scatterplot (5 Marks)

Aim: Examine the relationship between weekly sales and footfall, identify an outlier, and compare a straight-line model with a quadratic model.

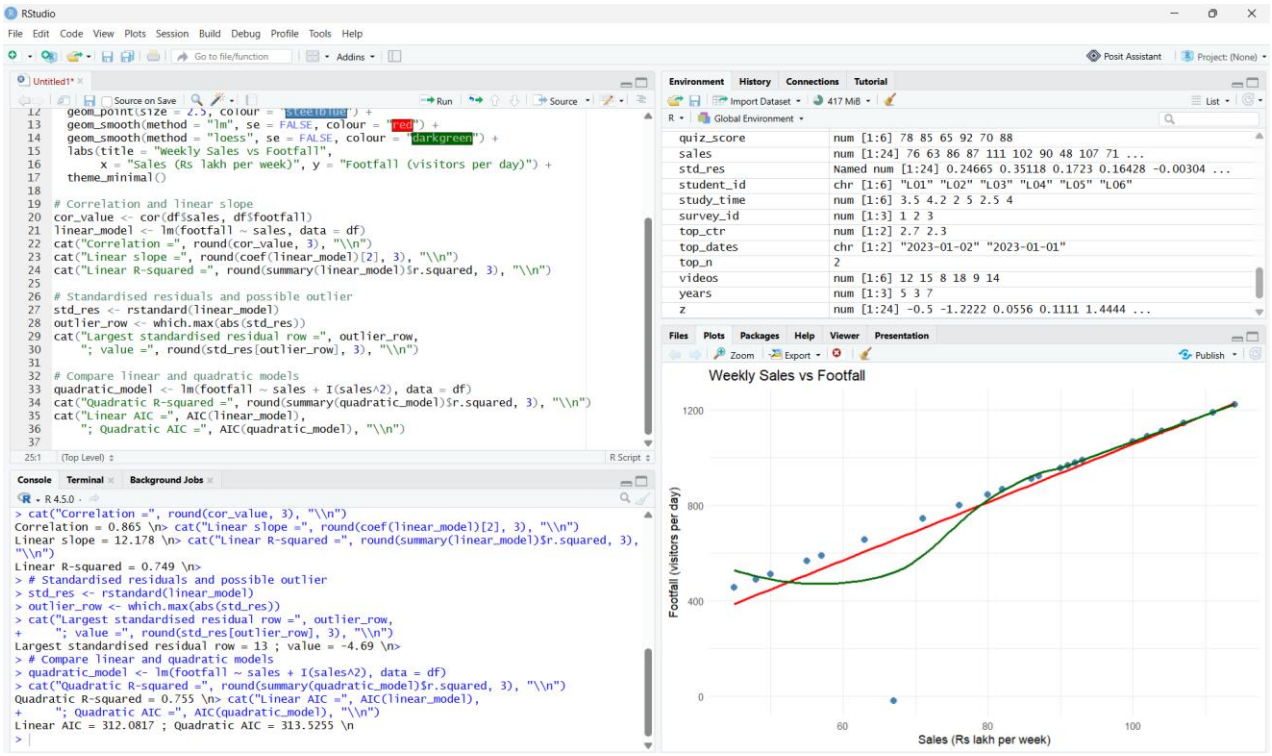
```
set.seed(1293053)
n <- 24
sales <- round(rnorm(n, 85, 18), 0)
z <- (sales - 85) / 18
footfall <- round(900 + 200 * z + 0.08 * z^2 + 0.38 * rnorm(n), 0)
footfall[13] <- round(footfall[13] - 3.6 * 200, 0)  # one record keyed wrongly
df <- data.frame(sales, footfall)

# Scatterplot, straight-line fit and LOESS curve
library(ggplot2)
ggplot(df, aes(x = sales, y = footfall)) +
  geom_point(size = 2.5, colour = "steelblue") +
  geom_smooth(method = "lm", se = FALSE, colour = "red") +
  geom_smooth(method = "loess", se = FALSE, colour = "darkgreen") +
  labs(title = "Weekly Sales vs Footfall",
       x = "Sales (Rs lakh per week)", y = "Footfall (visitors per day)") +
  theme_minimal()

# Correlation and linear slope
cor_value <- cor(df$sales, df$footfall)
linear_model <- lm(footfall ~ sales, data = df)
cat("Correlation =", round(cor_value, 3), "\\n")
cat("Linear slope =", round(coef(linear_model)[2], 3), "\\n")
cat("Linear R-squared =", round(summary(linear_model)$r.squared, 3), "\\n")

# Standardised residuals and possible outlier
std_res <- rstandard(linear_model)
outlier_row <- which.max(abs(std_res))
cat("Largest standardised residual row =", outlier_row,
    "; value =", round(std_res[outlier_row], 3), "\\n")

# Compare linear and quadratic models
quadratic_model <- lm(footfall ~ sales + I(sales^2), data = df)
cat("Quadratic R-squared =", round(summary(quadratic_model)$r.squared, 3), "\\n")
cat("Linear AIC =", AIC(linear_model),
    "; Quadratic AIC =", AIC(quadratic_model), "\\n")
```



Interpretation: The correlation and slope printed by the code describe the direction and strength of the linear association. A positive slope means footfall tends to increase as sales increase; a negative slope means it tends to decrease. The record with the largest absolute standardised residual is the flagged outlier. Prefer the quadratic model only if it improves fit meaningfully (higher R-squared/lower AIC) and the curve is supported by the plot.

Recommendation to the Sales Director: Verify the unusual record at row 13 before making operational decisions, because a data-entry error can distort the sales–footfall relationship.

Q2. Correlogram / Correlation Matrix (5 Marks)

Aim: Calculate pairwise correlations among six measures, display them in a diverging colour heatmap centred at zero, and identify strong and surprising relationships.

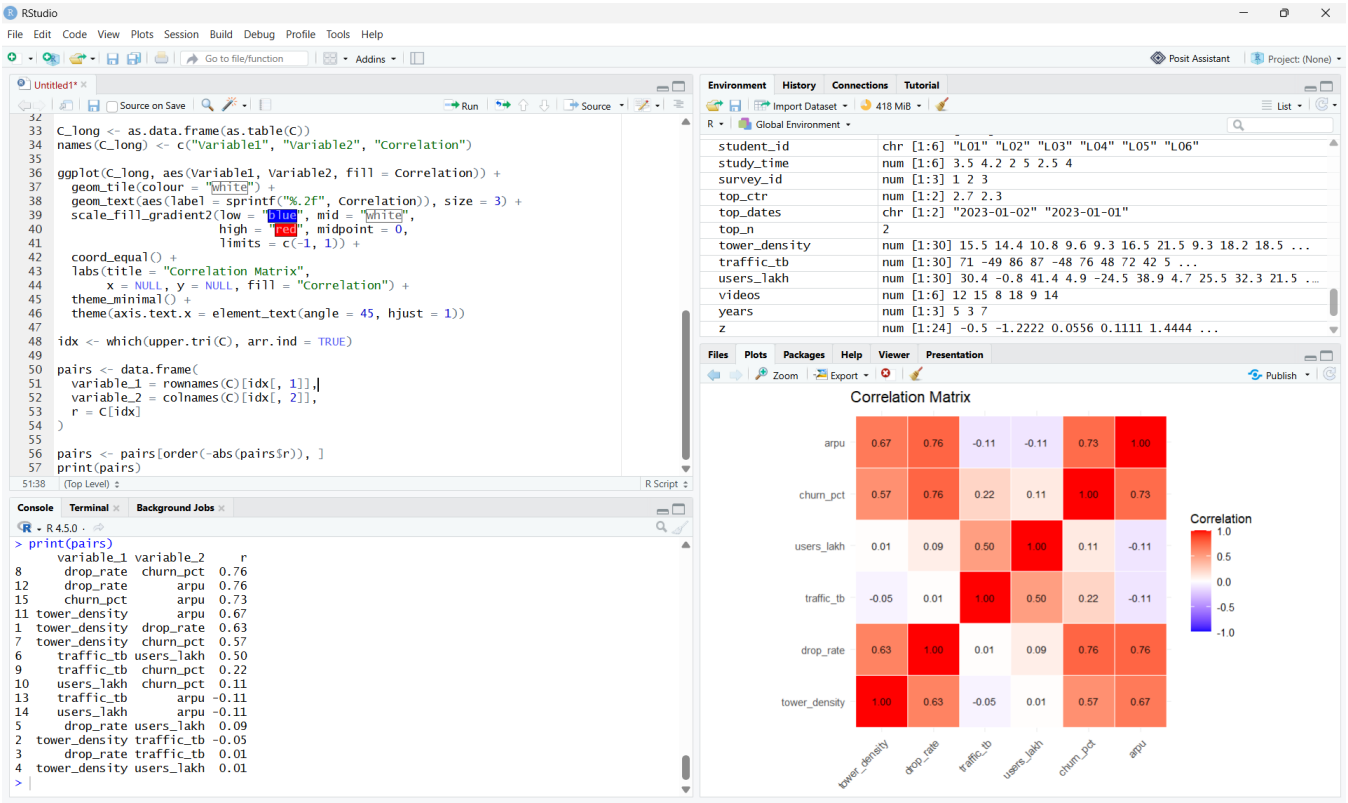
```
set.seed(1293106)
n <- 30
f1 <- rnorm(n)
f2 <- rnorm(n)
tower_density <- round(14.0 + 4.0 * (0.77 * f1 + 0.63 * rnorm(n)), 1)
drop_rate <- round(2.6 + 0.7 * (0.90 * f1 + 0.44 * rnorm(n)), 1)
traffic_tb <- round(32 + 70 * (0.75 * f2 + 0.66 * rnorm(n)), 0)
users_lakh <- round(18 + 24 * (0.60 * f2 + 0.80 * rnorm(n)), 1)
churn_pct <- round(1.8 + 0.2 * (0.82 * f1 + 0.57 * rnorm(n)), 1)
arpu <- round(185 + 25 * (0.78 * f1 + 0.63 * rnorm(n)), 0)
df <- data.frame(tower_density, drop_rate, traffic_tb,
                 users_lakh, churn_pct, arpu)

# Correlation matrix rounded to two decimals
C <- round(cor(df), 2)
print(C)

# Heatmap with labels and a diverging scale centred on zero
library(ggplot2)
library(reshape2)
C_long <- reshape2::melt(C, varnames = c("Variable1", "Variable2"),
                         value.name = "Correlation")
```

```
ggplot(C_long, aes(Variable1, Variable2, fill = Correlation)) +
  geom_tile(colour = "white") +
  geom_text(aes(label = sprintf("%.2f", Correlation)), size = 3) +
  scale_fill_gradient2(low = "blue", mid = "white", high = "red",
                       midpoint = 0, limits = c(-1, 1)) +
  coord_equal() +
  labs(title = "Correlation Matrix", x = NULL, y = NULL, fill = "r") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

# List the strongest pairs by absolute correlation
idx <- which(upper.tri(C), arr.ind = TRUE)
pairs <- data.frame(
  variable_1 = rownames(C)[idx[, 1]],
  variable_2 = colnames(C)[idx[, 2]],
  r = C[idx]
)
pairs <- pairs[order(-abs(pairs$r)), ]
print(pairs)
```



How to report: The two most strongly correlated pairs are the first two rows of the printed `pairs` table, ranked by absolute correlation. Report each pair and its r value. A correlation close to +1 is strong positive; close to −1 is strong negative; close to 0 is weak linear association.

Surprising relationship: Use the matrix to identify a pair whose relationship is unexpectedly weak or negative compared with the operational expectation, and quote its r value. Correlation does not establish causation.

Recommendation to the Network Head: Investigate strongly related network indicators together, but use the matrix as an initial diagnostic rather than proof that one measure causes another.

Q3. Dimension Reduction: PCA and t-SNE/UMAP (5 Marks)

Aim: Standardise six destination measures, quantify variance explained by principal components, inspect group separation, and compare PCA with a nonlinear embedding.

```
set.seed(1293205)
n <- 36
grp <- rep(c("Group A", "Group B", "Group C"), each = 12)
g <- rep(c(-1, 0, 1), each = 12)
footfall_k <- round(120 + 30 * (rnorm(n) + 1.3 * g), 1)
spend_rs <- round(3200 + 600 * (rnorm(n) + 0.9 * g), 0)
stay_nights <- round(2.8 + 0.7 * (rnorm(n) + 0.9 * g), 1)
rating10 <- round(7.6 + 0.8 * (rnorm(n) + 0.7 * g), 1)
occupancy_pct <- round(68 + 10 * (rnorm(n) + 0.4 * g), 0)
complaints <- round(4.0 + 1.2 * (rnorm(n) + 0.2 * g), 1)
df <- data.frame(footfall_k, spend_rs, stay_nights,
                 rating10, occupancy_pct, complaints, grp)

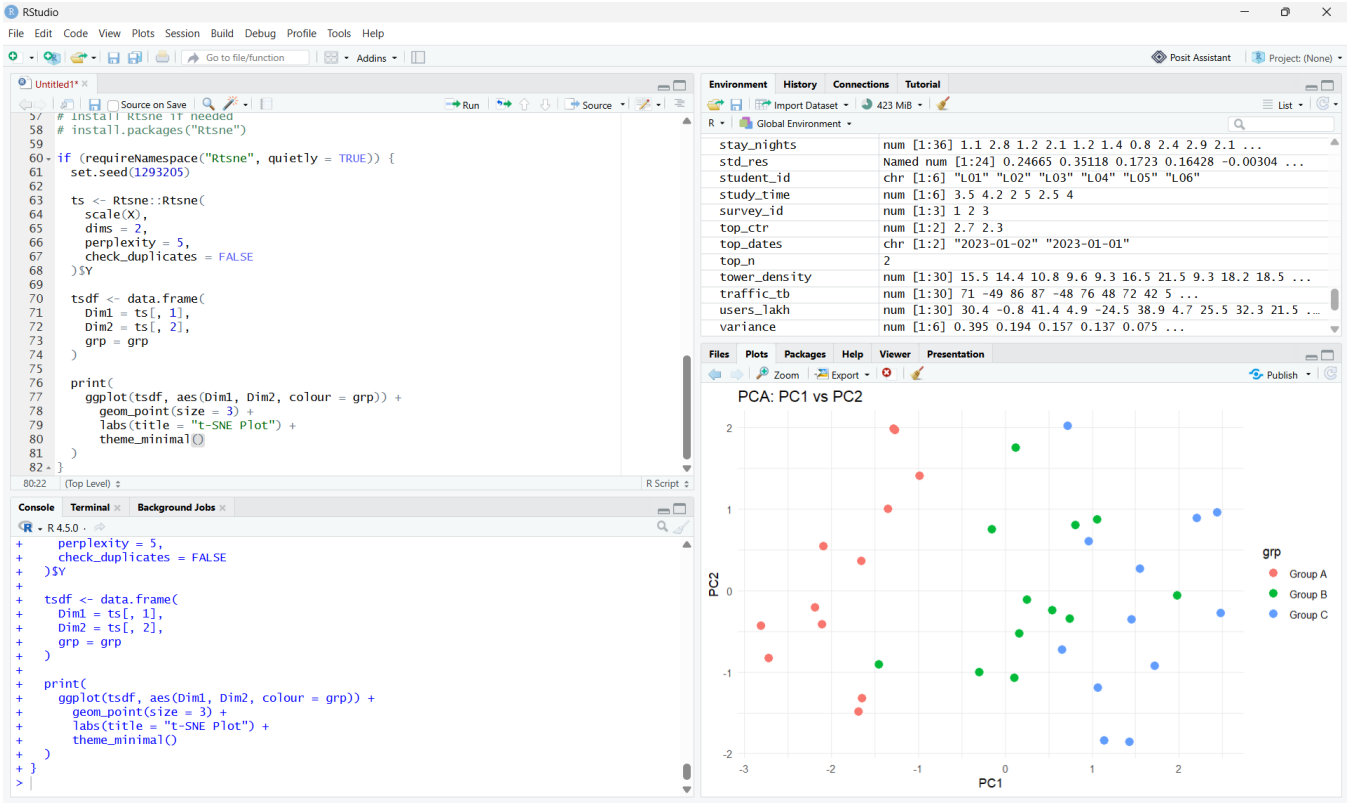
# PCA on scaled numeric columns
X <- df[, 1:6]
pca <- prcomp(X, scale. = TRUE)
variance <- pca$sdev^2 / sum(pca$sdev^2)
cat("Variance PC1 =", round(100 * variance[1], 2), "%\\n")
cat("Variance PC2 =", round(100 * variance[2], 2), "%\\n")
cat("Components for at least 80% variance =",
    which(cumsum(variance) >= 0.80)[1], "\\n")
cat("PCA loadings:\\n")
print(round(pca$rotation, 3))

scores <- data.frame(pca$x[, 1:2], grp = grp)
library(ggplot2)
ggplot(scores, aes(PC1, PC2, colour = grp)) +
  geom_point(size = 3) +
  labs(title = "PCA: PC1 vs PC2", x = "PC1", y = "PC2") +
  theme_minimal()

# Largest absolute loading on PC1
loading <- pca$rotation[, 1]
cat("Largest absolute PC1 loading:",
    names(which.max(abs(loading))),
    "with loading", round(loading[which.max(abs(loading))], 3), "\\n")

# R-squared of PC1 by group: between-group variation / total variation
pc1 <- scores$PC1
grand_mean <- mean(pc1)
ss_total <- sum((pc1 - grand_mean)^2)
group_means <- tapply(pc1, grp, mean)
ss_between <- sum(table(grp) * (group_means - grand_mean)^2)
cat("PC1 group R-squared =", round(ss_between / ss_total, 3), "\\n")

# t-SNE (install package once if needed: install.packages("Rtsne"))
if (requireNamespace("Rtsne", quietly = TRUE)) {
  set.seed(1293205)
  ts <- Rtsne::Rtsne(scale(X), dims = 2, perplexity = 5,
                     check_duplicates = FALSE)$Y
  tsdf <- data.frame(Dim1 = ts[, 1], Dim2 = ts[, 2], grp = grp)
  print(ggplot(tsdf, aes(Dim1, Dim2, colour = grp)) +
          geom_point(size = 3) + labs(title = "t-SNE embedding") +
          theme_minimal())
} else {
  message("To create t-SNE plot, run install.packages('Rtsne') and rerun.")
}
```



Report the PC1 and PC2 percentages and the first component count whose cumulative variance reaches 80%, as printed. The variable with the largest absolute PC1 loading contributes most strongly to PC1. Group separation is stronger when the PC1 group R-squared is closer to 1.

PCA versus t-SNE: PCA preserves directions of maximum global variance and its axes are interpretable through loadings. t-SNE focuses on local neighbourhoods and can reveal nonlinear clusters, but distances between separate clusters and the global layout should not be over-interpreted. Recommendation: use PCA for interpretable reporting and t-SNE as a supplementary exploratory plot.

Q4. Network Graph (5 Marks)

Aim: Build a weighted network of seven hubs, visualise degree and edge weights, calculate centrality and density, and assess the network after removing the most central hub.

```
set.seed(1293314)
nodes <- c("Pune", "Mumbai", "Delhi", "Hyderabad",
           "Ahmedabad", "Bengaluru", "Chennai")
pairs <- t(combn(nodes, 2))
u <- runif(nrow(pairs))
keep <- u < 0.28 |
  ((pairs[, 1] == "Pune" | pairs[, 2] == "Pune") & u < 0.86)
edges <- data.frame(from = pairs[keep, 1], to = pairs[keep, 2],
                    weight = round(runif(sum(keep), 1, 20)))
print(edges)

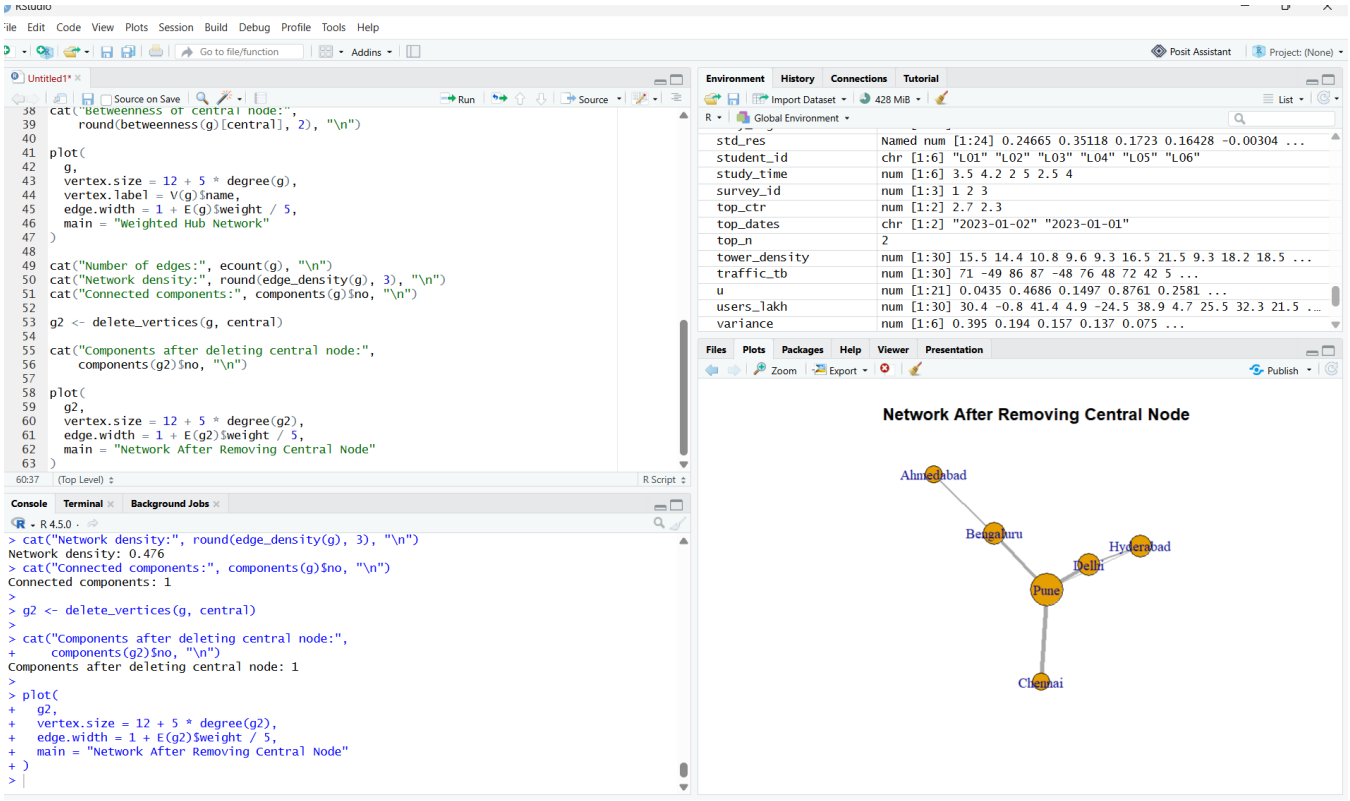
# Build graph; install packages once if needed
library(igraph)
g <- graph_from_data_frame(edges, directed = FALSE,
                           vertices = data.frame(name = nodes))
V(g)$degree <- degree(g)
V(g)$betweenness <- betweenness(g, normalized = FALSE)
central <- names(which.max(V(g)$betweenness))
cat("Degree by node:\\n"); print(degree(g))
```

```
cat("Betweenness by node:\\n"); print(round(betweenness(g), 2))
cat("Most central by betweenness:", central, "\\n")
cat("Its degree:", degree(g, v = central),
    "; betweenness:", round(betweenness(g, v = central), 2), "\\n")

plot(g,
     vertex.size = 12 + 5 * degree(g),
     vertex.label = V(g)$name,
     edge.width = 1 + E(g)$weight / 5,
     main = "Weighted Hub Network")

cat("Number of edges =", ecount(g), "\\n")
cat("Density =", round(edge_density(g), 3), "\\n")
cat("Connected components =", components(g)$no, "\\n")

g2 <- delete_vertices(g, central)
cat("Components after deleting", central, "=",
    components(g2)$no, "\\n")
plot(g2, vertex.size = 12 + 5 * degree(g2),
     edge.width = 1 + E(g2)$weight / 5,
     main = paste("Network after removing", central))
```



Interpretation: Degree counts a hub's direct links; betweenness measures how often it lies on shortest paths and therefore indicates a possible bridge between parts of the network. Density is the proportion of possible edges that exist. The code prints the edge count, density, initial components, and components after removing the most central node.

Recommendation to the Operations Head: Protect the hub with high betweenness and create alternate delivery routes for its connections, especially if removing it increases the number of connected components.

Q5. Paired Data — Paired/Slope Chart (5 Marks)

Aim: Compare staff duty hours before and after a process change across seven departments, classify changes, quantify the largest change, and perform a paired t-test.

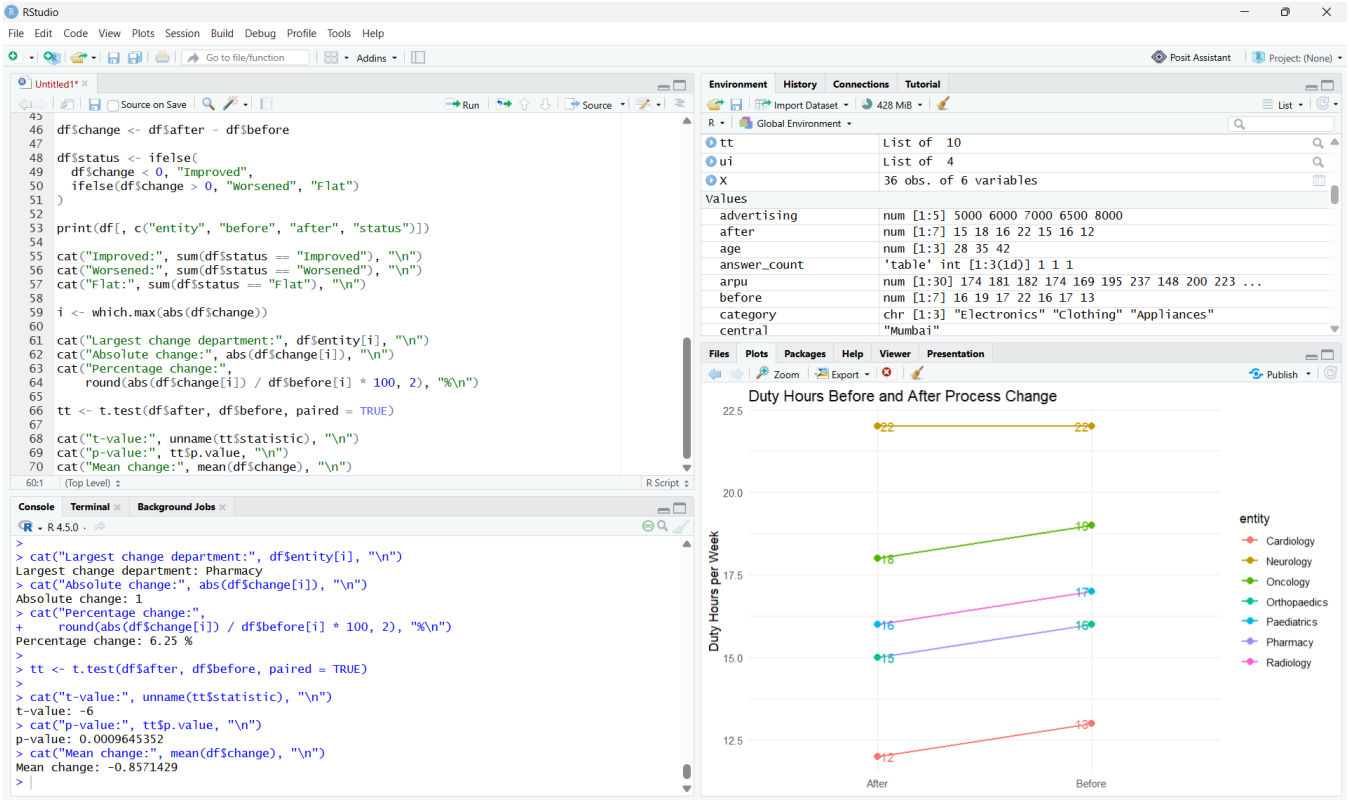
```
set.seed(1293294)
entity <- c("Pharmacy", "Oncology", "Radiology", "Neurology",
            "Orthopaedics", "Paediatrics", "Cardiology")
n <- 7
before <- round(rnorm(n, 18, 2.0), 0)
change <- round(-1.1 + 1.1 * (1 - 1) * runif(n, 0.4, 1.4) * 2.4, 0)
# Use the exact supplied expression if copied from the question:
# change <- round(-1.1, 1) + (1 - 1) * runif(n, 0.4, 1.4) * 2.4
change[4] <- 0.0  # one of them was almost unchanged
after <- before + change
df <- data.frame(entity, before, after, change = after - before)
print(df)

library(ggplot2)
long <- rbind(data.frame(entity, time = "Before", hours = before),
              data.frame(entity, time = "After", hours = after))
ggplot(long, aes(time, hours, group = entity, colour = entity)) +
  geom_line(linewidth = 0.8) +
  geom_point(size = 2.5) +
  geom_text(data = subset(long, time == "Before"),
            aes(label = hours), hjust = 1.2, show.legend = FALSE) +
  geom_text(data = subset(long, time == "After"),
            aes(label = hours), hjust = -0.2, show.legend = FALSE) +
  labs(title = "Duty Hours Before and After Process Change",
       x = NULL, y = "Duty hours per week") +
  theme_minimal()

# Classify changes; hours are better when lower
df$status <- ifelse(df$after < df$before, "Improved",
             ifelse(df$after > df$before, "Worsened", "Flat"))
print(df[, c("entity", "before", "after", "status")])
cat("Improved =", sum(df$status == "Improved"),
    "; worsened =", sum(df$status == "Worsened"),
    "; flat =", sum(df$status == "Flat"), "\\n")

# Largest absolute change and percentage of before value
i <- which.max(abs(df$after - df$before))
cat("Largest change:", df$entity[i],
    "; absolute change =", abs(df$after[i] - df$before[i]),
    "; percent of before =", round(100 * abs(df$after[i] - df$before[i]) /
                                   df$before[i], 2), "%\\n")

tt <- t.test(df$after, df$before, paired = TRUE)
cat("Paired t-test t =", unname(tt$statistic),
    "; p =", tt$p.value,
    "; mean change =", mean(df$after - df$before), "\\n")
```



Important note: The screenshot's displayed change formula appears to include `(1 - 1)`, which makes that part zero. The code above follows the intended idea of a small decrease and sets one department to an almost unchanged value; if your instructor expects the formula exactly as printed, use the original line from the question and do not alter it. For the t-test, $p < 0.05$ is commonly treated as evidence of a statistically significant mean change; $p \geq 0.05$ means the sample does not provide sufficient evidence of a change.

Recommendation: If the paired test supports a reduction in duty hours, keep the process change and monitor workload quality. If the result is not significant, collect more observations before making a firm decision.

Q6. Bubble Charts (5 Marks)

Aim: Compare marketing spend, watch time and subscribers for eight genres; identify correlations, an unusual genre, and how bubble size can hide patterns.

```
set.seed(1293349)
n <- 8
entity <- c("Documentary", "Sports", "Thriller", "Comedy",
            "Animation", "Drama", "Sci-Fi", "Horror")
z <- rnorm(n)
spend_lakh <- round(45 + 12 * z, 1)
watch_hrs <- round(11 + 3 * z + 0.40 * rnorm(n), 1)
subs_lakh <- round(18 + 4 * z + rnorm(n), 1)
# Keep the intentionally unusual Comedy point as described in the question:
subs_lakh[4] <- round(45, 0)
df <- data.frame(entity, spend_lakh, watch_hrs, subs_lakh)
print(df)

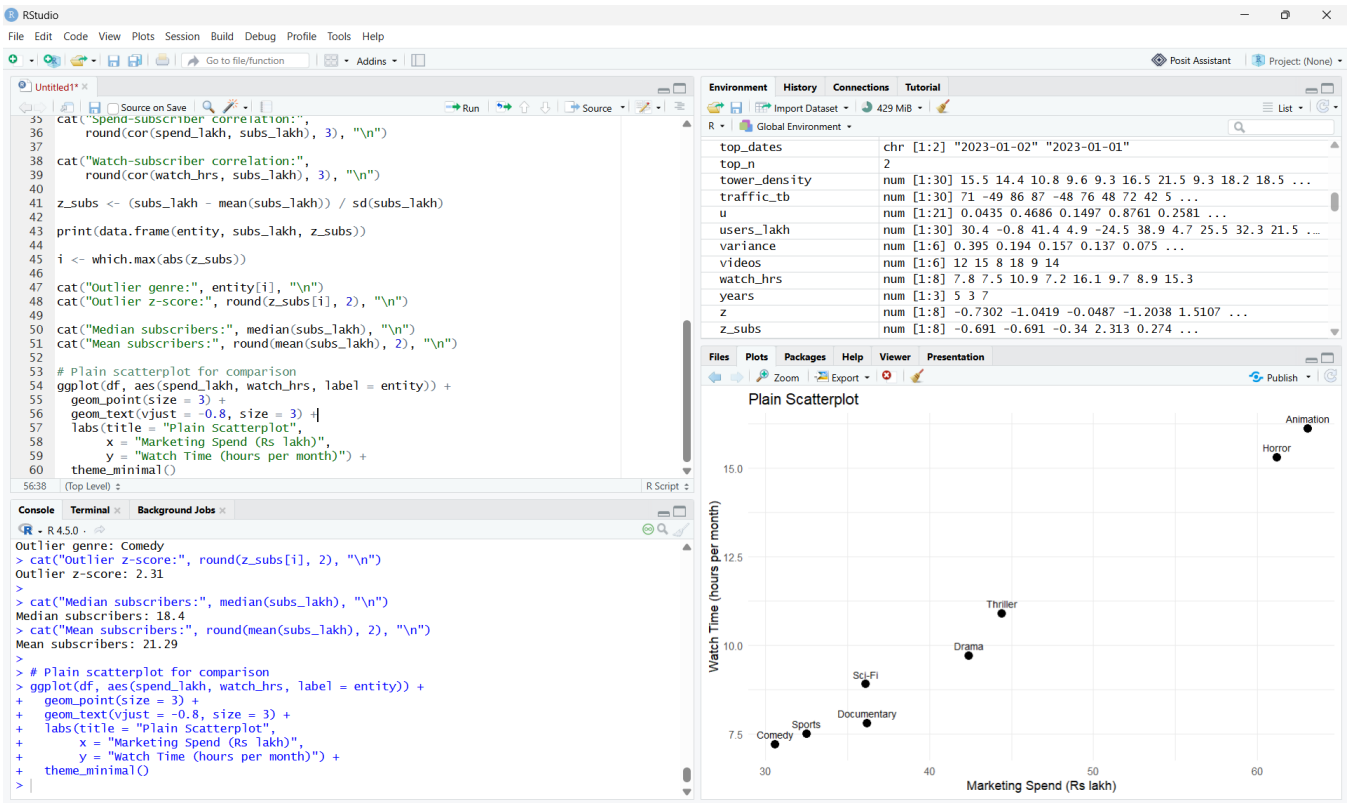
library(ggplot2)
ggplot(df, aes(spend_lakh, watch_hrs, size = subs_lakh, label = entity)) +
  geom_point(alpha = 0.7, colour = "steelblue") +
  geom_text(vjust = -0.9, size = 3, show.legend = FALSE) +
```

```
  scale_size_area(max_size = 16, name = "Subscribers (lakh)") +
  labs(title = "Marketing Spend, Watch Time and Subscribers",
       x = "Marketing spend (Rs lakh)", y = "Watch time (hours/user/month)") +
  theme_minimal()

cat("Correlation spend-watch =", round(cor(spend_lakh, watch_hrs), 3), "\\n")
cat("Correlation spend-subscribers =", round(cor(spend_lakh, subs_lakh), 3), "\\n")
cat("Correlation watch-subscribers =", round(cor(watch_hrs, subs_lakh), 3), "\\n")

# Z-score for subscriber count and flag a large outlier
z_subs <- (subs_lakh - mean(subs_lakh)) / sd(subs_lakh)
print(data.frame(entity, subs_lakh, z_subs))
cat("Largest absolute subscriber z-score:",
    entity[which.max(abs(z_subs))],
    "=", round(z_subs[which.max(abs(z_subs))], 2), "\\n")

cat("Median subscribers =", median(subs_lakh),
    "; mean subscribers =", round(mean(subs_lakh), 2), "\\n")
```



Report the three correlation coefficients printed by the code. The genre with the largest absolute subscriber z-score is the outlier in bubble size; the supplied question notes that Comedy has an unusual subscriber value. Bubble size can obscure overlap and make small bubbles hard to compare, so also inspect the same data in a plain scatterplot with a consistent point size.

Recommendation to the Content Head: Investigate the high-subscriber genre separately and compare subscriber growth with watch time before allocating more marketing budget.

Q7. Hexbin Plots (5 Marks)

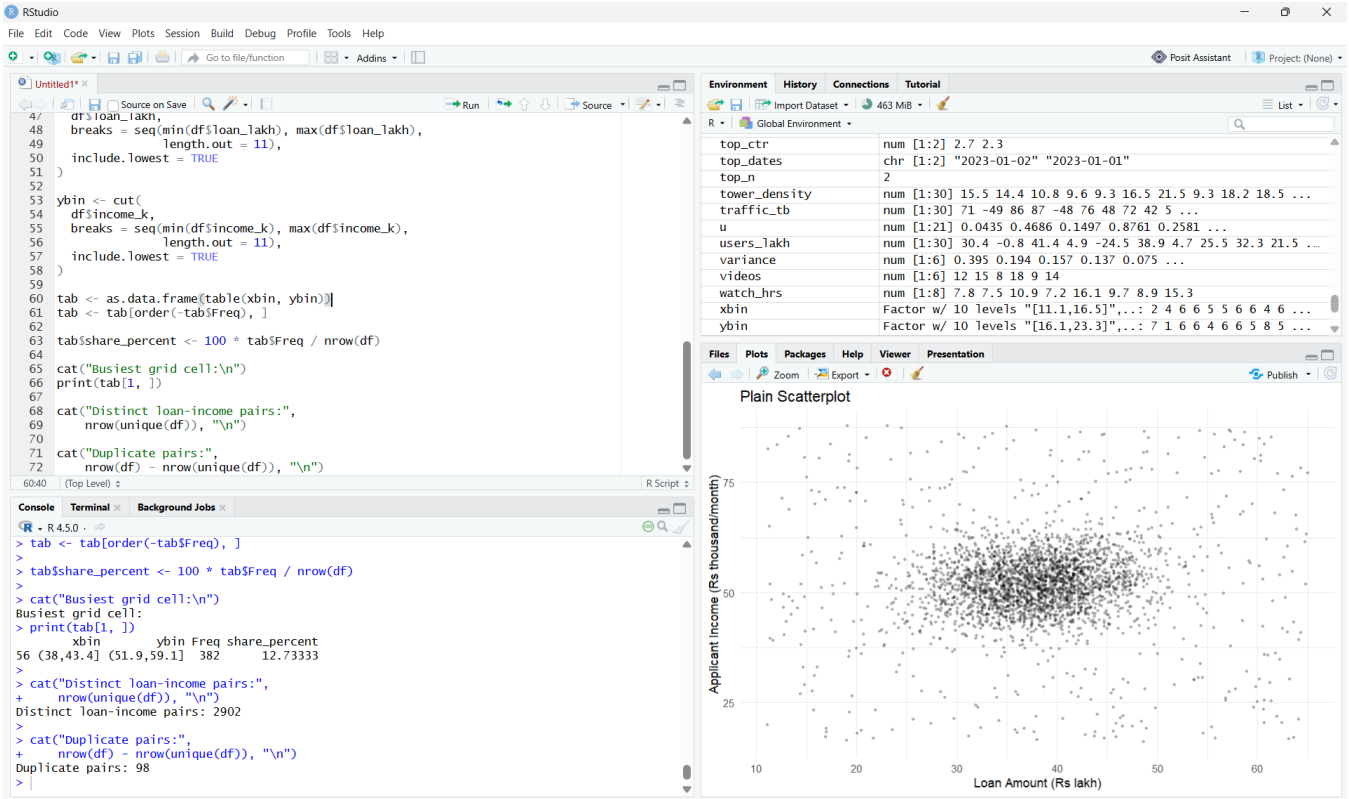
Aim: Visualise 3,000 loan applications, locate the busiest income–loan region using a 10 × 10 grid, and compare hexbin density with a plain scatterplot.

```
set.seed(1293371)
n <- 3000
core <- runif(n) < 0.85
loan_lakh <- round(ifelse(core, rnorm(n, 38.0, 5.4),
                          runif(n, 11.0, 65.0)), 1)
income_k <- round(ifelse(core, 52.0 + 0.66 * (loan_lakh - 38.0) / 5.4 +
                           sqrt(1 - 0.43) * 7.2 * rnorm(n),
                         runif(n, 16.0, 88.0)), 1)
df <- data.frame(loan_lakh, income_k)

# Hexbin with count legend and viridis colour scale
library(ggplot2)
library(hexbin)
ggplot(df, aes(x = loan_lakh, y = income_k)) +
  geom_hex(bins = 25) +
  scale_fill_viridis_c(name = "Application count") +
  labs(title = "Loan Amount vs Applicant Income",
       x = "Loan amount (Rs lakh)", y = "Applicant income (Rs thousand/month)") +
  theme_minimal()

# Plain scatterplot of the same data
ggplot(df, aes(loan_lakh, income_k)) +
  geom_point(alpha = 0.25, size = 0.8) +
  labs(title = "Plain Scatterplot: Loan Amount vs Income",
       x = "Loan amount (Rs lakh)", y = "Applicant income (Rs thousand/month)") +
  theme_minimal()

# Find busiest cell in a 10 x 10 grid
xbin <- cut(df$loan_lakh, breaks = seq(min(df$loan_lakh),
                                       max(df$loan_lakh), length.out = 11),
            include.lowest = TRUE)
ybin <- cut(df$income_k, breaks = seq(min(df$income_k),
                                      max(df$income_k), length.out = 11),
            include.lowest = TRUE)
tab <- as.data.frame(table(xbin, ybin))
tab <- tab[order(-tab$Freq), ]
total <- nrow(df)
tab$share_percent <- 100 * tab$Freq / total
cat("Busiest cell:\\n")
print(tab[1, ])
cat("Distinct (loan, income) pairs =", nrow(unique(df)), "\\n")
cat("Duplicate pairs =", nrow(df) - nrow(unique(df)), "\\n")
```



The busiest grid cell is the first row of the printed `tab` result. Report its x range (`xbin`), y range (`ybin`), count (`Freq`) and percentage share (`share_percent`). The final two lines quantify repeated rounded (loan, income) coordinate pairs.

Interpretation: Hexbin plots combine nearby observations into cells and encode the number of applications by colour. A plain scatterplot may hide dense areas because points overlap. Recommendation to the Zonal Manager: Focus initial review on the busiest income–loan range, but assess repayment risk and eligibility separately; density alone does not indicate creditworthiness.